



14B CHEMICAL KINETICS

UNIT 02: HOW REACTIONS HAPPEN

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 14

1. Express reactions rates in relation to reaction stoichiometry.
2. Use experimental data to write rate laws for chemical reactions.
3. Predict the concentration and time values of reactions using first- and second-order integrated rate laws.
4. Graph data to identify the order of a reaction.
5. Define and apply the terms half-life, first-order reaction, and second-order reaction.
6. Explain collision theory using appropriate terminology.
7. Apply the Arrhenius equation to calculate activation energy and/or the value of the rate constant at various temperatures.
8. Define and discuss reaction mechanisms.
9. Write rate laws for elementary steps.
10. Identify intermediates in a reaction mechanism.
11. Discuss how catalysts work to speed up chemical reactions

Reaction Rates and Concentration: Rate Laws

14.2

The initial rate is the *instantaneous* rate *as close as possible* to the *start* of the reaction (“time zero”).

- Reactions are generally *fastest* at the beginning, and slow down thereafter.
- The use of initial rates permits us to ignore the concentrations of *products* during chemical reactions.
- Chemists often measure the initial (instantaneous) rate of the reactants.

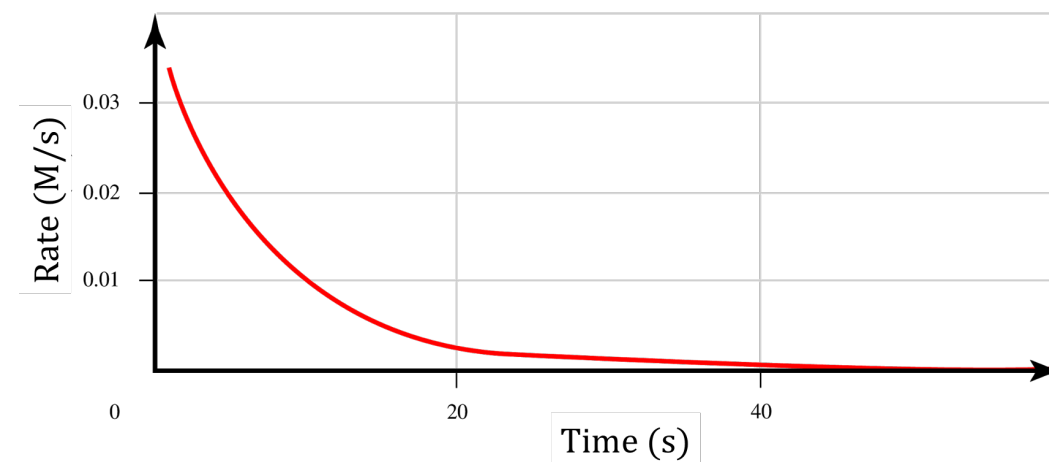


FIGURE 14.7 Instantaneous rate of a reaction over time

From this point onward, any time we use the term “*rate*” we mean the initial rate.

Reaction Rates and Concentration: Rate Laws

14.2

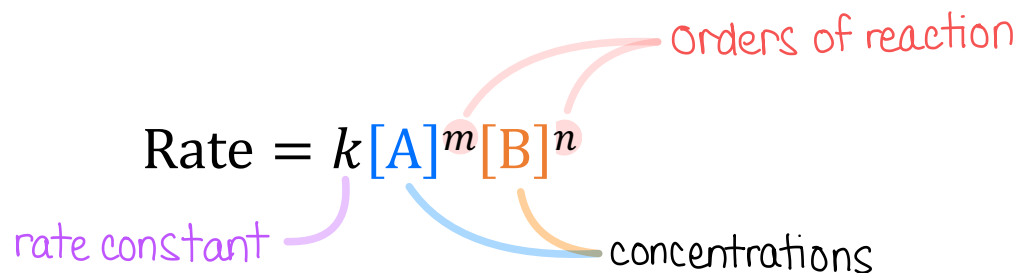
Rate law

A mathematical equation that summarizes the dependence of a reaction's rate on the concentrations of the reactants.

There are two useful forms of the rate law:

- 1) Differential rate laws (usually referred to simply as the rate law)
- 2) Integrated rate laws

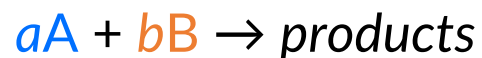
For a generic reaction: $aA + bB \rightarrow \text{products}$



Reaction Rates and Concentration: Rate Laws

14.2

For a generic reaction:



$$\text{Rate} = k[A]^m[B]^n$$

rate constant

concentrations

Orders

Rate constant (k)

- A proportionality constant relating the rate to the concentrations of the reactants.
- The value of k is specific to a reaction at a particular temperature.

Orders of reaction

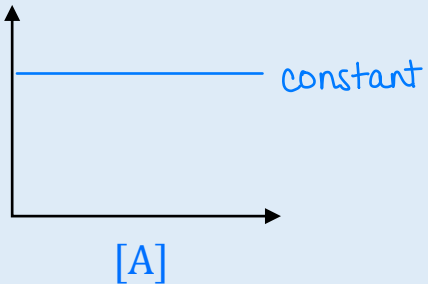
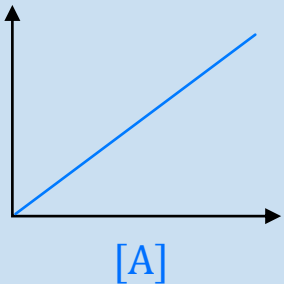
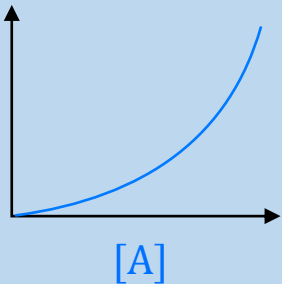
- Represented by powers m and n in the example above.
- The orders define how the rate is affected by the concentration of each reactant.
- Often, orders are integers or half-integers, that must be determined experimentally.

Reaction Rates and Concentration: Rate Laws

14.2

For a generic reaction with a single reactant A : $aA \rightarrow \text{products}$

$$\text{Rate} = k[A]^m$$

	Zeroth Order ($m = 0$)	First Order ($m = 1$)	Second Order ($m = 2$)
Rate $\propto [A]$	Rate does not depend on $[A]$ $[A] \times 2 \Rightarrow \text{Rate} \times 1$ $[A] \times 3 \Rightarrow \text{Rate} \times 1$ $[A] \times \frac{1}{2} \Rightarrow \text{Rate} \times 1$	Rate depends linearly on $[A]$ $[A] \times 2 \Rightarrow \text{Rate} \times 2$ $[A] \times 3 \Rightarrow \text{Rate} \times 3$ $[A] \times \frac{1}{2} \Rightarrow \text{Rate} \times \frac{1}{2}$	Rate depends quadratically on $[A]$ $[A] \times 2 \Rightarrow \text{Rate} \times 4$ $[A] \times 3 \Rightarrow \text{Rate} \times 9$ $[A] \times \frac{1}{2} \Rightarrow \text{Rate} \times \frac{1}{4}$
Form	$\text{Rate} = k[A]^0 = k$	$\text{Rate} = k[A]^1$	$\text{Rate} = k[A]^2$
Plot	Rate  $[A]$	Rate  $[A]$	Rate  $[A]$

Reaction Rates and Concentration: Rate Laws

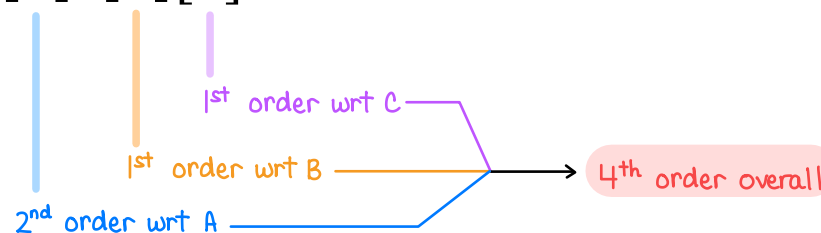
14.2

Many reactions have *more than one* reactant.

- Each reactant has its own order of reaction.
- We say that each is the “order **with respect to**” a particular reactant.
- The overall order of reaction is the *sum* of the orders of each reactant.

Example) What is the overall order of reaction for a chemical reaction with the following rate law?

$$\text{Rate} = k[A]^2[B][C]$$



Reaction Rates and Concentration: Rate Laws

14.2

Units for the rate constant (k) depend on the overall order ($\mathcal{O}$) of the reaction.

$$\text{Rate} = k[A]^m[B]^n \dots \quad \mathcal{O} = m + n + \dots$$

Suppose the rates are measured in $\frac{\text{M}}{\text{s}}$.

The units for k will *generally* be:

$$\text{M}^{1-\mathcal{O}} \cdot \text{s}^{-1}$$

Overall Order ($\mathcal{O}$)		Units of k		
Zeroth	$\mathcal{O} = 0$	$\frac{\text{M}}{\text{s}}$	or	$\text{M} \cdot \text{s}^{-1}$
First	$\mathcal{O} = 1$	$\frac{1}{\text{s}}$	or	s^{-1}
Second	$\mathcal{O} = 2$	$\frac{1}{\text{M} \cdot \text{s}}$	or	$\text{M}^{-1} \cdot \text{s}^{-1}$
Third	$\mathcal{O} = 3$	$\frac{1}{\text{M}^2 \cdot \text{s}}$	or	$\text{M}^{-2} \cdot \text{s}^{-1}$
Fourth	$\mathcal{O} = 4$	$\frac{1}{\text{M}^3 \cdot \text{s}}$	or	$\text{M}^{-3} \cdot \text{s}^{-1}$

Reaction Rates and Concentration: Rate Laws

14.2

The method of initial rates is the experimental method used to determine the orders of a reaction with respect to each reactant.

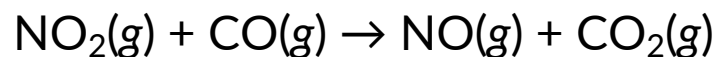
It involves these general steps:

- 1) Start with a base (or *standard* or *reference*) set of concentrations for each reactant and measure the initial rate of this reaction (experiment).
- 2) Change the concentration of *only one reactant at a time* to isolate its impact on the rate.
- 3) Compare the rates of the two experiments to deduce the order with respect to the reactant chosen.

EXAMPLE PROBLEM

Learning Objective(s): 14.2

Determine the rate law expression and the rate constant (k) for the following reaction given the tabulated experimental data obtained at 298 K.



Expt.	$[\text{NO}_2]_{\text{initial}}$	$[\text{CO}]_{\text{initial}}$	Initial Rate
1	0.263 M	0.826 M	$1.44 \times 10^{-5} \text{ M/s}$
2	0.263 M	0.413 M	$1.44 \times 10^{-5} \text{ M/s}$
3	0.526 M	0.413 M	$5.76 \times 10^{-5} \text{ M/s}$

Step 1
General form: $\text{Rate} = k \cdot [\text{NO}_2]^m [\text{CO}]^n$

Step 2

For m , compare expts. 2 and 3
because $[\text{CO}]$ is constant,
so only $[\text{NO}_2]$ changing.

$$\frac{\text{Rate}_3}{\text{Rate}_2} = \frac{k \cdot [\text{NO}_2]_3^m [\text{CO}]_3^n}{k \cdot [\text{NO}_2]_2^m [\text{CO}]_2^n}$$

$$\frac{5.76 \times 10^{-5} \text{ M/s}}{1.44 \times 10^{-5} \text{ M/s}} = \left(\frac{0.526 \text{ M}}{0.263 \text{ M}} \right)^m$$

$$4 = 2^m$$

$$m = 2$$

Step 3

For n , compare expts. 1 and 2
because $[\text{NO}_2]$ is constant,
so only $[\text{CO}]$ changing.

$$\frac{\text{Rate}_1}{\text{Rate}_2} = \frac{[\text{CO}]_1^n}{[\text{CO}]_2^n}$$

$$\frac{1.44 \times 10^{-5} \text{ M/s}}{1.44 \times 10^{-5} \text{ M/s}} = \left(\frac{0.826 \text{ M}}{0.413 \text{ M}} \right)^n$$

$$1 = 2^n$$

$$n = 0$$

Step 4

For k , pick any expt and
plug values into rate law.
Let's choose expt. 1!

$$\text{Rate} = k \cdot [\text{NO}_2]^2$$

$$1.44 \times 10^{-5} \frac{\text{M}}{\text{s}} = k \cdot (0.263 \text{ M})^2$$

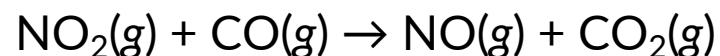
$$k = \frac{1.44 \times 10^{-5} \frac{\text{M}}{\text{s}}}{(0.263 \text{ M})^2}$$

$$k = 0.000208 \text{ M}^{-1} \cdot \text{s}^{-1}$$

IN-CLASS QUESTION 1

Learning Objective(s): 14.2

Calculate the rate of the reaction (in M/s) for experiment 4 in the table below.



Expt.	$[\text{NO}_2]_{\text{initial}}$	$[\text{CO}]_{\text{initial}}$	Initial Rate
1	0.263 M	0.826 M	$1.44 \times 10^{-5} \text{ M/s}$
2	0.263 M	0.413 M	$1.44 \times 10^{-5} \text{ M/s}$
3	0.526 M	0.413 M	$5.76 \times 10^{-5} \text{ M/s}$
4	0.789 M	1.239 M	?

The rate law and rate constant (k) can be used to predict rates.

$$\text{Rate} = k \cdot [\text{NO}_2]^2, \quad k = 0.000208 \text{ M}^{-1} \cdot \text{s}^{-1}$$
$$= (0.000208 \text{ M}^{-1} \cdot \text{s}^{-1}) \cdot (0.789 \text{ M})^2$$

$$\text{Rate} = 1.30 \times 10^{-4} \frac{\text{M}}{\text{s}}$$

Alternatively, rate law tells us that if $[\text{NO}_2]$ changes by some factor x , then the rate changes by x^2 !

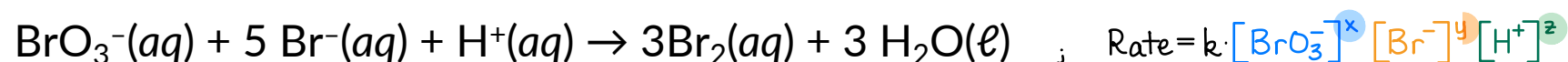
Comparing expts 4 to 1, $[\text{NO}_2]$ changed by 3, so rate must change by $3^2 = 9$!

$$\text{Rate}_4 = 3^2 \cdot \text{Rate}_1 = 1.30 \times 10^{-4} \frac{\text{M}}{\text{s}}$$

IN-CLASS QUESTION 2 & 3

Learning Objective(s): 14.2

Determine the rate law and rate constant for the following reaction using the given experimental data.



Expt.	$[\text{BrO}_3^-]_{\text{initial}}$	$[\text{Br}^-]_{\text{initial}}$	$[\text{H}^+]_{\text{initial}}$	Initial Rate
1	0.10 M	0.10 M	0.10 M	$8.0 \times 10^{-4} \text{ M/s}$
2	0.20 M	0.10 M	0.10 M	$1.6 \times 10^{-3} \text{ M/s}$
3	0.20 M	0.20 M	0.10 M	$3.2 \times 10^{-3} \text{ M/s}$
4	0.10 M	0.10 M	0.20 M	$3.2 \times 10^{-3} \text{ M/s}$

For k , pick any expt and plug values into rate law. Let's choose expt. 1!

$$\begin{aligned} \text{Rate}_1 &= k [\text{BrO}_3^-]_1^x [\text{Br}^-]_1^y [\text{H}^+]_1^z \\ 8.0 \times 10^{-4} \frac{\text{M}}{\text{s}} &= k (0.10 \text{ M})^x (0.10 \text{ M})^y (0.10 \text{ M})^z \\ k &= \frac{8.0 \times 10^{-4} \frac{\text{M}}{\text{s}}}{(0.10 \text{ M})^4} \\ k &= 8.0 \text{ M}^{-3} \cdot \text{s}^{-1} \end{aligned}$$

For x , compare expts. 1 and 2: $k, [\text{Br}^-], [\text{H}^+]$ constants

$$\frac{1.6 \times 10^{-3} \text{ M/s}}{8.0 \times 10^{-4} \text{ M/s}} = \left(\frac{0.20 \text{ M}}{0.10 \text{ M}} \right)^x$$

$x=1$

For y , compare expts. 2 and 3: $k, [\text{BrO}_3^-], [\text{H}^+]$ constants

$$\frac{3.2 \times 10^{-3} \text{ M/s}}{1.6 \times 10^{-3} \text{ M/s}} = \left(\frac{0.20 \text{ M}}{0.10 \text{ M}} \right)^y$$

$y=1$

For z , compare expts. 1 and 4: $k, [\text{BrO}_3^-], [\text{Br}^-]$ constants

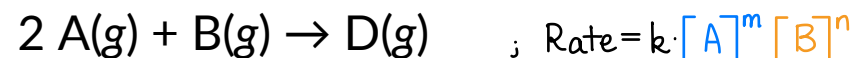
$$\frac{3.2 \times 10^{-3} \text{ M/s}}{8.0 \times 10^{-4} \text{ M/s}} = \left(\frac{0.20 \text{ M}}{0.10 \text{ M}} \right)^z$$

$z=2$

EXAMPLE PROBLEM

Learning Objective(s): 14.2

Determine the rate law expression and the rate constant (k) for the following reaction given the tabulated experimental data obtained at 25 °C.



Expt.	[A] _{initial}	[B] _{initial}	Initial Rate
1	0.20 M	0.30 M	$6.0 \times 10^{-4} \text{ M/s}$
2	0.20 M	0.10 M	$2.0 \times 10^{-4} \text{ M/s}$
3	0.60 M	0.40 M	$7.2 \times 10^{-3} \text{ M/s}$

For n , compare expts. 1 and 2:

$$\frac{\text{Rate}_1}{\text{Rate}_2} = \frac{k \cdot [\text{A}]_1^m [\text{B}]_1^n}{k \cdot [\text{A}]_2^m [\text{B}]_2^n} \quad ; \quad k \text{ and } [\text{A}] \text{ constants}$$

$$\frac{6.0 \times 10^{-4} \text{ M/s}}{2.0 \times 10^{-4} \text{ M/s}} = \left(\frac{0.30 \text{ M}}{0.10 \text{ M}} \right)^n$$

$$3 = 3^n$$

$$n = 1$$

EXAMPLE PROBLEM

Learning Objective(s): 14.2

Determine the rate law expression and the rate constant (k) for the following reaction given the tabulated experimental data obtained at 25 °C.



Expt.	[A] _{initial}	[B] _{initial}	Initial Rate
1	0.20 M	0.30 M	$6.0 \times 10^{-4} \text{ M/s}$
2	0.20 M	0.10 M	$2.0 \times 10^{-4} \text{ M/s}$
3	0.60 M	0.40 M	$7.2 \times 10^{-3} \text{ M/s}$

For m , no two expts. have $[\text{B}]$ constant but already know $n=1$, so no need to isolate $[\text{A}]$.
Pick expts. 1+3 or 2+3 to compare.

$$\frac{\text{Rate}_3}{\text{Rate}_2} = \frac{k \cdot [\text{A}]_3^m [\text{B}]_3^1}{k \cdot [\text{A}]_2^m [\text{B}]_2^1} \quad ; \quad k \text{ constant}$$

$$\frac{7.2 \times 10^{-3} \text{ M/s}}{2.0 \times 10^{-4} \text{ M/s}} = \left(\frac{0.60 \text{ M}}{0.20 \text{ M}} \right)^m \left(\frac{0.40 \text{ M}}{0.10 \text{ M}} \right)^1$$

$$36 = 3^m \cdot 4$$

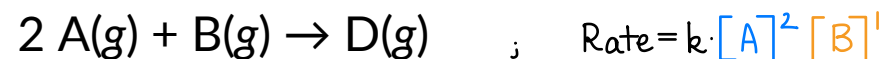
$$9 = 3^m$$

$$m = 2$$

EXAMPLE PROBLEM

Learning Objective(s): 14.2

Determine the rate law expression and the rate constant (k) for the following reaction given the tabulated experimental data obtained at 25 °C.



Expt.	[A] _{initial}	[B] _{initial}	Initial Rate
1	0.20 M	0.30 M	$6.0 \times 10^{-4} \text{ M/s}$
2	0.20 M	0.10 M	$2.0 \times 10^{-4} \text{ M/s}$
3	0.60 M	0.40 M	$7.2 \times 10^{-3} \text{ M/s}$

For k , pick any expt. Let's choose expt. 2!

$$\text{Rate}_2 = k \cdot [\text{A}]_2^2 \cdot [\text{B}]_2^1$$

$$2.0 \times 10^{-4} \frac{\text{M}}{\text{s}} = k \cdot (0.20 \text{ M})^2 \cdot (0.10 \text{ M})^1$$
$$k = \frac{2.0 \times 10^{-4} \frac{\text{M}}{\text{s}}}{(0.20 \text{ M})^2 \cdot (0.10 \text{ M})^1}$$

$$k = 0.050 \text{ M}^{-2} \cdot \text{s}^{-1}$$

BONUS QUESTION

Assessed on: Fall 2021, Exam 3

Determine the rate law for the reaction given the initial rates data in the table below.



☐ Rate = $k[\text{NO}]^2[\text{O}_2]^2$

☐ Rate = $k[\text{NO}][\text{O}_2]$

☐ Rate = $k[\text{NO}]^2[\text{O}_2]$

☐ Rate = $k[\text{NO}][\text{O}_2]^2$

☐ Rate = k

Expt.	$[\text{NO}]_{\text{initial}}$	$[\text{O}_2]_{\text{initial}}$	Initial Rate
1	0.0422 M	0.113 M	$2.35 \times 10^{-5} \text{ M/s}$
2	0.0844 M	0.113 M	$9.40 \times 10^{-5} \text{ M/s}$
3	0.0844 M	0.226 M	$1.88 \times 10^{-4} \text{ M/s}$